

Code for Model

User properties

#	^	Name	Type	Value	Unit
1	[/]	tox_offset	Length	0.65	um
2	[/]	target_radius	Length	25	um

```
# =====  
# 1. FIBER & PORT 1  
# =====
```

```
fiber_xpos = getnamed("fiber","x");  
fiber_ypos = getnamed("fiber","y");  
fiber_zpos = getnamed("fiber","z");  
fiber_core_diameter = getnamed("fiber","core diameter");  
fiber_core_index = getnamed("fiber","core index");  
fiber_theta = getnamed("fiber","theta");
```

```
tox_zpos = getnamed("SiO2 TOX","z");
```

```
fiber_theta_rad = fiber_theta*(pi/180.0);  
port_xpos = fiber_xpos + tox_offset* sin(fiber_theta_rad);  
port_zpos = tox_zpos + fiber_zpos + tox_offset* cos(fiber_theta_rad);  
setnamed("FDTD:ports:port 1","x",port_xpos);  
setnamed("FDTD:ports:port 1","y",fiber_ypos);  
setnamed("FDTD:ports:port 1","z",port_zpos);
```

```
# Port angle and offset  
setnamed("FDTD:ports:port 1","theta",fiber_theta);  
port_offset = 4.0*fiber_core_diameter*tan(fiber_theta_rad);  
setnamed("FDTD:ports:port 1","rotation offset",port_offset);
```

```
# =====  
# 2. RADIUS, PORT 2 & REGION  
# =====
```

```
setnamed("grating","radius", target_radius);
```

```
# Get the grating's center X position  
grating_x = getnamed("grating", "x");
```

```
# Calculate the end of the taper, then pull Port 2 back 1um to the left  
wg_start_x = grating_x - target_radius;  
port2_x = wg_start_x - 1e-6;
```

```
# Move Port 2  
setnamed("FDTD:ports:port 2", "x", port2_x);
```

```
# Dynamically stretch the FDTD simulation boundaries  
# 2um buffer behind Port 2, 12um buffer past the grating center  
fdtd_xmin = port2_x - 2e-6;  
fdtd_xmax = grating_x + 12e-6;
```

```
setnamed("FDTD","x min", fdtd_xmin);  
setnamed("FDTD","x max", fdtd_xmax);
```

```
# =====  
# 3. MONITOR  
# =====
```

```
# Index Monitor  
setnamed("monitor","x min", fdtd_xmin);  
setnamed("monitor","x max", fdtd_xmax);
```

```
# Movie Monitor  
setnamed("monitor_1","x min", fdtd_xmin);  
setnamed("monitor_1","x max", fdtd_xmax);
```

Code for Grating

User properties

#	^	Name	Type	Value	Unit
1		target length	Length	25	um
2		h total	Length	1	um
3		etch depth	Length	0.3	um
4		duty cycle	Number	0.5	
5		pitch	Length	0.84	um
6		radius	Length	25	um
7		y span	Length	10	um
8		L extra	Length	10	um
9		waveguide width	Length	0.3	um
10		index	Number	3.5	
11		material	Material	Si3N4 - 1000nm	
12		m	Number	1.15	
13		waveguide length	Length	10	um

```
deleteall;

n_periods = ceil(target length%/pitch);
fill_width = pitch*%duty cycle%;
etch_width = pitch*(1-%duty cycle%);
L = n_periods*pitch + etch_width;
theta = asin( 0.5*%y span%/radius ) * 180/pi;

# waveguide
addrect;
set("name","waveguide");
set("x min",-%waveguide length%);
set("x max",0);
set("y",0);
set("y span",%waveguide width%);
set("z min",0);
set("z max",%h total%);

# input section
adding;
set("name","total section");
set("inner radius",0);
neg = -(%waveguide width%/2) * tan( (90 - theta) * pi/180 );
set("x",neg);
set("y",0);
set("z min",0);
set("z max",%h total%);
set("theta start",-theta);
set("theta stop",theta);

set("outer radius", radius + abs(neg));

# lower layer below grating
if(%etch depth% < %h total%) {
    adding;
    set("name","lower layer");
    neg = -(%waveguide width%/2) * tan( (90 - theta) * pi/180 );
    set("x",neg);
    set("y",0);
    set("theta start",-theta);
    set("theta stop",theta);
    set("inner radius",radius);
    set("outer radius",radius+L);
    set("z min",0);
    set("z max",%h total%-%etch depth%);
}

# output section
adding;
set("name","output section");
neg = -(%waveguide width%/2) * tan( (90 - theta) * pi/180 );

set("x",neg);
set("y",0);
set("theta start",-theta);
set("theta stop",theta);
set("inner radius",radius+L);
set("outer radius",radius+L+%L extra%);
set("z min",0);
set("z max",%h total%);

#add grating
for(i=1:n_periods){
    adding;
    neg = -(%waveguide width%/2) * tan( (90 - theta) * pi/180 );

    set("x",neg);
    set("y",0);
    set("z min",%h total%-%etch depth%);
    set("z max",%h total%);
    set("theta start",-theta);
    set("theta stop",theta);
    set("inner radius",radius + pitch*(i-1)+etch_width);
    set("outer radius",radius + pitch*i);
}

selectall;
set("material",material);
if(get("material")=="<Object defined dielectric>")
{ set("index",index); }
move(-radius,0,0);
```

Code for Fiber

User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	12	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

```
# materials
core_index = %core index%;
cladding_index = %cladding index%;

# dimensions
core_radius = %core diameter%/2.0;
cladding_radius = %cladding diameter%/2.0;

# core
theta_rad = theta/(180/pi);
L = %z span%/cos(theta_rad);
setnamed("core","radius",core_radius);
setnamed("core","index",core_index);
setnamed("core","x",0.0);
setnamed("core","y",0.0);
setnamed("core","z",0.0);
setnamed("core","z span",L);
setnamed("core","first axis","y");
setnamed("core","rotation 1",theta);

# cladding
setnamed("cladding","radius",cladding_radius);
setnamed("cladding","index",cladding_index);
setnamed("cladding","x",0.0);
setnamed("cladding","y",0.0);
setnamed("cladding","z",0.0);
setnamed("cladding","z span",L);
setnamed("cladding","first axis","y");
setnamed("cladding","rotation 1",theta);
```

Code for Other layers

 Edit rectangle

name

Geometry Material Rotations Graphical rendering

x (μm)	0	x min (μm)	-60
x span (μm)	120	x max (μm)	60

y (μm)	0	y min (μm)	-60
y span (μm)	120	y max (μm)	60

z (μm)	0.75	z min (μm)	0
z span (μm)	1.5	z max (μm)	1.5

☒ use relative coordinates

 Edit rectangle

name

Geometry Material Rotations Graphical rendering

x (μm)	0	x min (μm)	-60
x span (μm)	120	x max (μm)	60

y (μm)	0	y min (μm)	-60
y span (μm)	120	y max (μm)	60

z (μm)	-1	z min (μm)	-2
z span (μm)	2	z max (μm)	0

☒ use relative coordinates

 Edit rectangle

name

Geometry Material Rotations Graphical rendering

x (μm)	0	x min (μm)	-60
x span (μm)	120	x max (μm)	60

y (μm)	0	y min (μm)	-60
y span (μm)	120	y max (μm)	60

z (μm)	-3	z min (μm)	-4
z span (μm)	2	z max (μm)	-2

☒ use relative coordinates

Code for FDTD and MONITORS

Edit FDTD simulation

name

General

Geometry

Mesh settings

Boundary conditions

Advanced o

x (μm)

x span (μm)

x min (μm)

x max (μm)

y (μm)

y span (μm)

y min (μm)

y max (μm)

z (μm)

z span (μm)

z min (μm)

z max (μm)

Edit port

name

Geometry

Modal Properties

Impedance

x (μm)

x span (μm)

x min (μm)

x max (μm)

y (μm)

y span (μm)

y min (μm)

y max (μm)

z (μm)

z span (μm)

z min (μm)

z max (μm)

☐ use relative coordinates

Edit port

name

Geometry

Modal Properties

Impedance

x (μm)

x span (μm)

x min (μm)

x max (μm)

y (μm)

y span (μm)

y min (μm)

y max (μm)

z (μm)

z span (μm)

z min (μm)

z max (μm)

☐ use relative coordinates

Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	0.25	um
3	[/]	etch depth	Length	VARIABLE	um
4	[]	duty cycle	Number	0.4	
5	[/]	pitch	Length	0.689	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	15	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	2	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Fiber Property

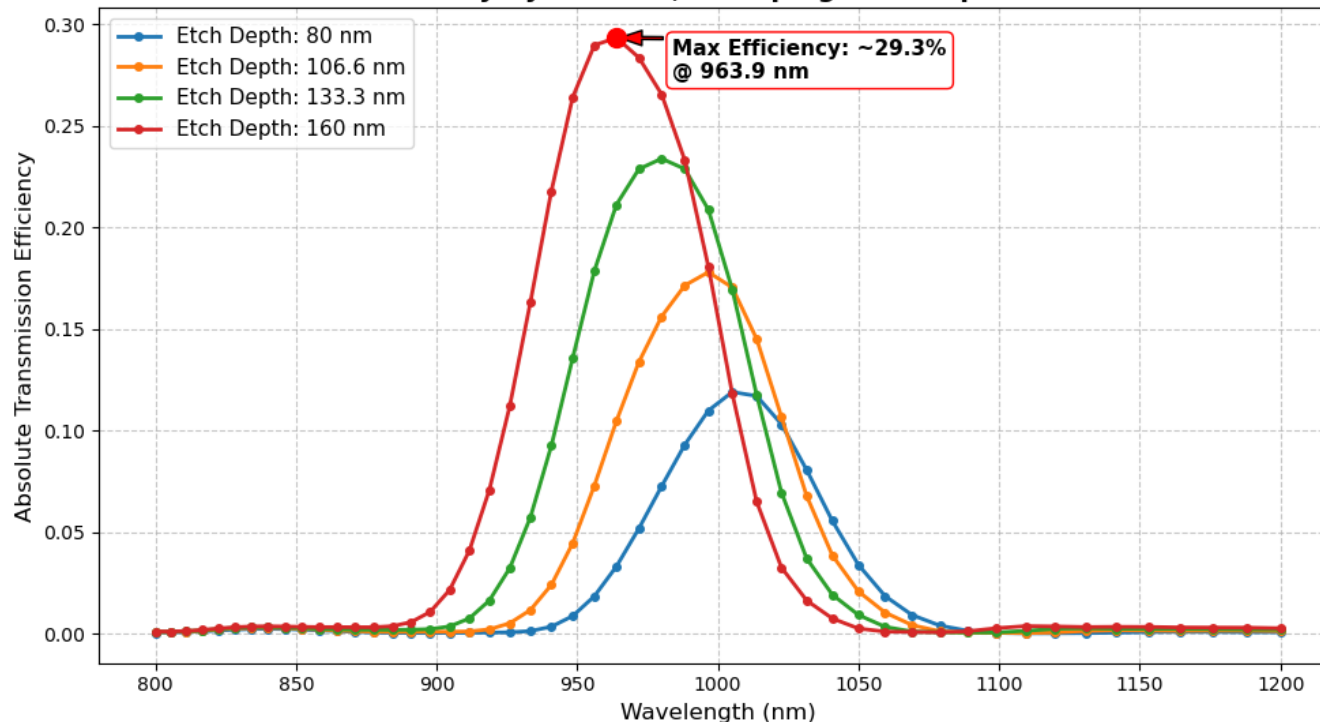
User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	8	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

Result :-Transmission 29.3% at 963.9_{nm}

Etching Depth Variable from 0.08_{um} to 0.160_{um}
Fixing Duty Cycle at 0.4
Box layer SiO₂ = 1.206_{um}

Grating Efficiency vs. Wavelength
(Duty Cycle = 0.4, Sweeping Etch Depth)



Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	0.25	um
3	[/]	etch depth	Length	VARIABLE	um
4	[]	duty cycle	Number	0.5	
5	[/]	pitch	Length	0.689	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	15	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	2	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Fiber Property

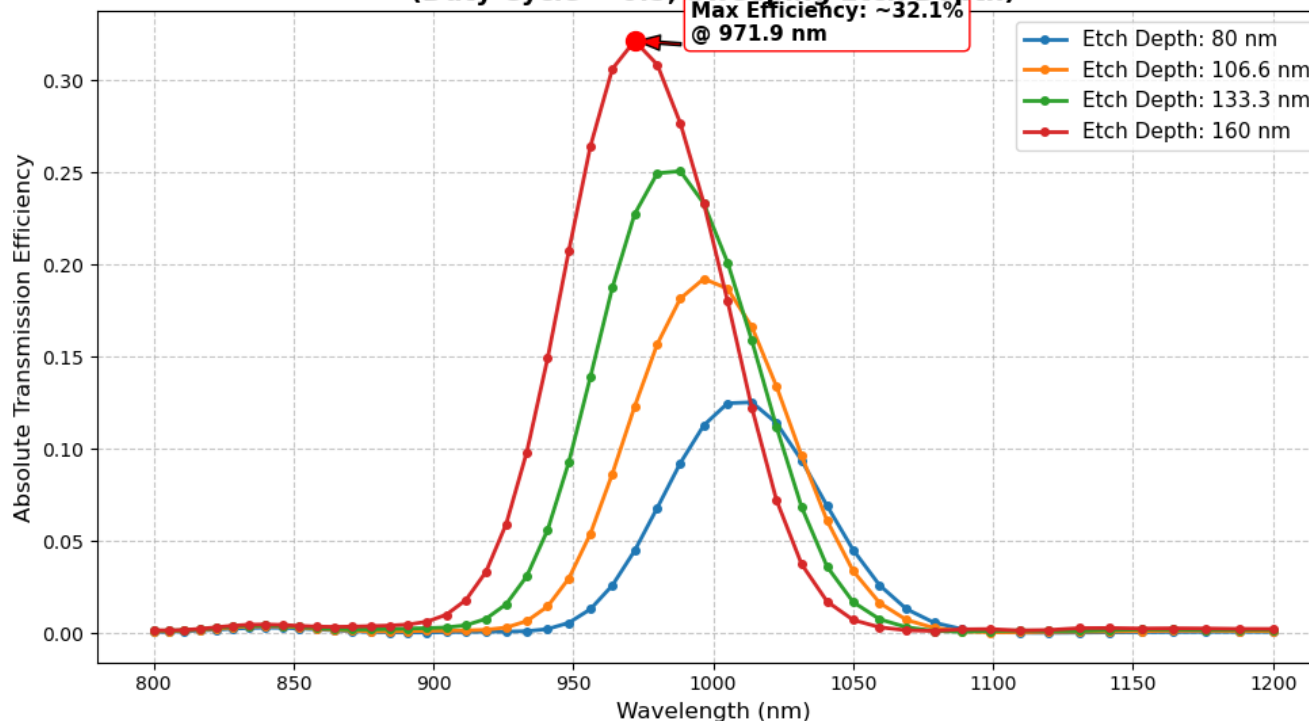
User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	8	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

Result :-Transmission 32% at 971.9_{nm}
Increase From Before, Wavelength shifted to Right

Etching Depth Variable from 0.08_{um} to 0.160_{um}
Fixing Duty Cycle at 0.5
Box layer SiO₂ = 1.206_{um}

Grating Efficiency vs. Wavelength
(Duty Cycle = 0.5, Sweeping Etch Depth)



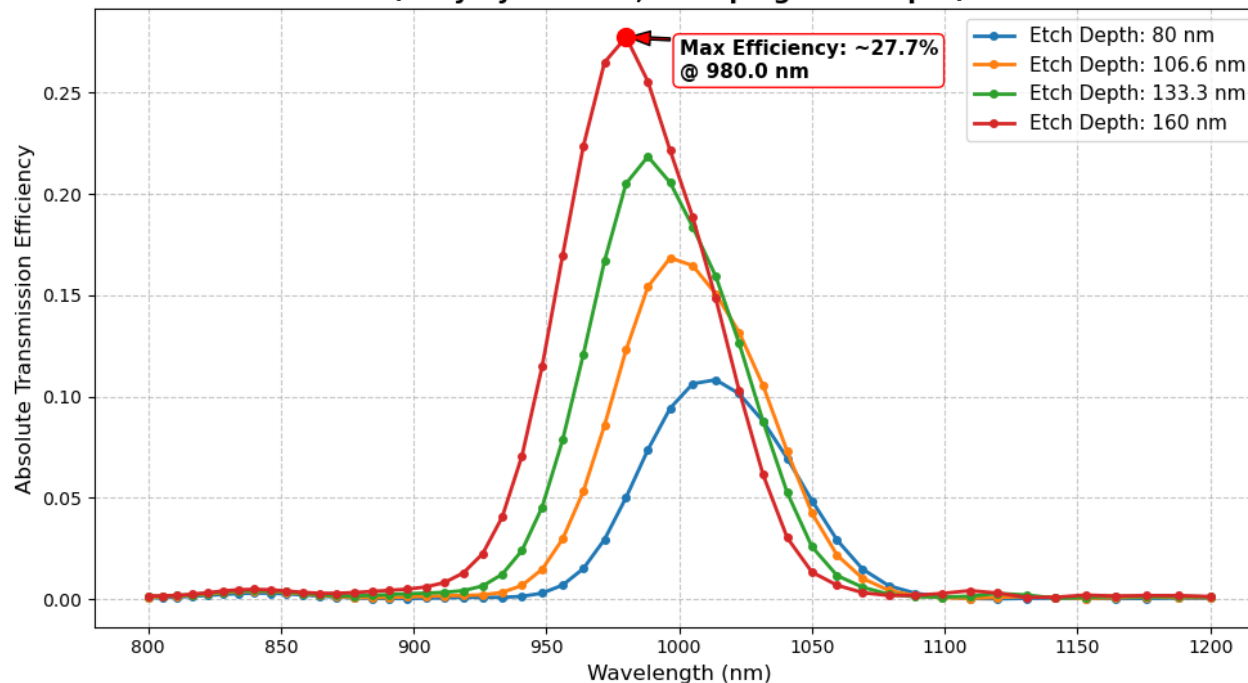
Etching Depth Variable from $0.08_{\mu\text{m}}$ to $0.160_{\mu\text{m}}$
 Fixing Duty Cycle at 0.6
 Box layer $\text{SiO}_2 = 1.206_{\mu\text{m}}$

Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	0.25	um
3	[/]	etch depth	Length	VARIABLE	um
4	[]	duty cycle	Number	0.6	
5	[/]	pitch	Length	0.689	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	15	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	2	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Grating Efficiency vs. Wavelength
 (Duty Cycle = 0.6, Sweeping Etch Depth)



Fiber Property

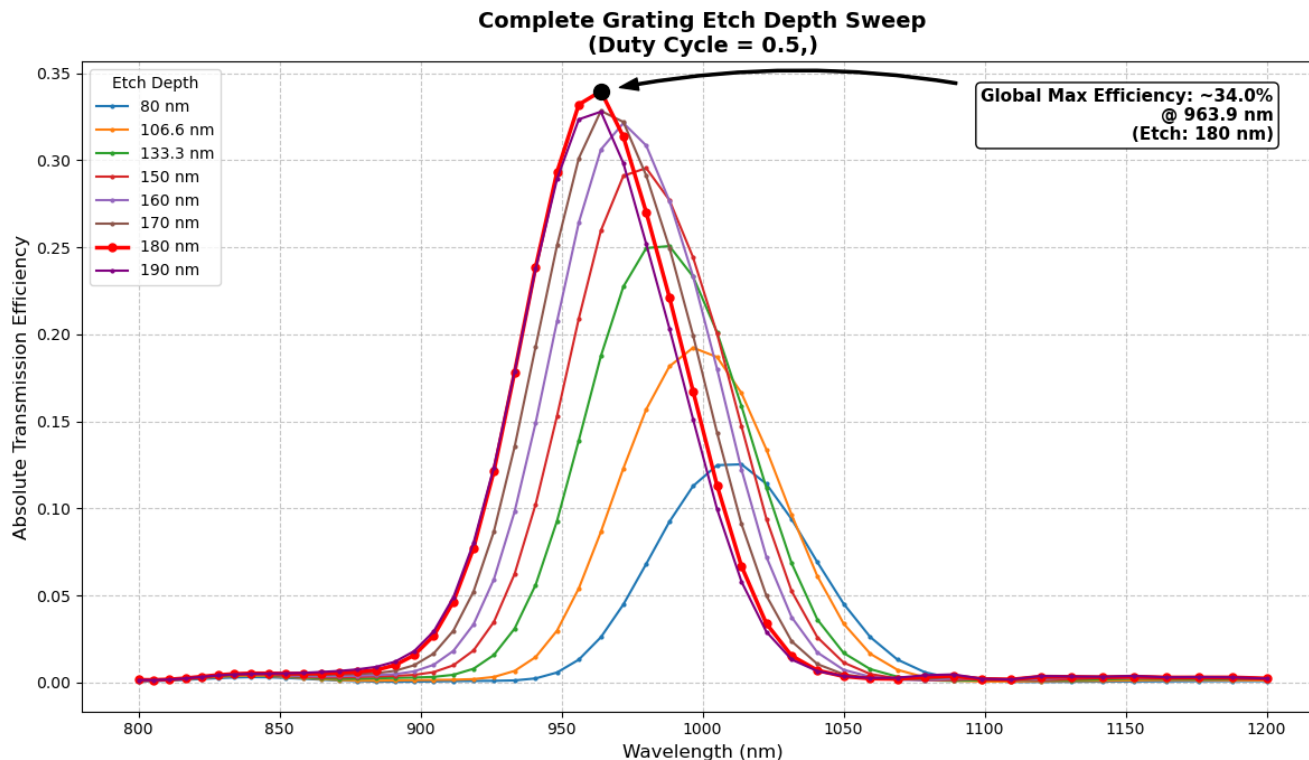
User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	8	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

Result :-Transmission 27.7% at 980_{nm}
Decrease From Before, Wavelength shifted to Right

Duty Cycle 0.5 Gives the Height Efficient among theme

Etching Depth Variable from $0.08_{\mu\text{m}}$ to $0.190_{\mu\text{m}}$
Fixing Duty Cycle at 0.5
Box layer $\text{SiO}_2 = 1.206_{\mu\text{m}}$



Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	0.25	um
3	[/]	etch depth	Length	VARIABLE	um
4	[]	duty cycle	Number	0.5	
5	[/]	pitch	Length	0.689	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	15	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	3.5	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Fiber Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	8	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

Height Transmission 34% at 963.9_{nm} at
Etching Depth $0.180_{\mu\text{m}}$
Duty Cycle at 0.5

Fibre Angle From 8 to 12

Etching Depth $0.180_{\mu\text{m}}$
Duty Cycle at 0.5

Box layer $\text{SiO}_2 = 1.206_{\mu\text{m}}$

Grating Property

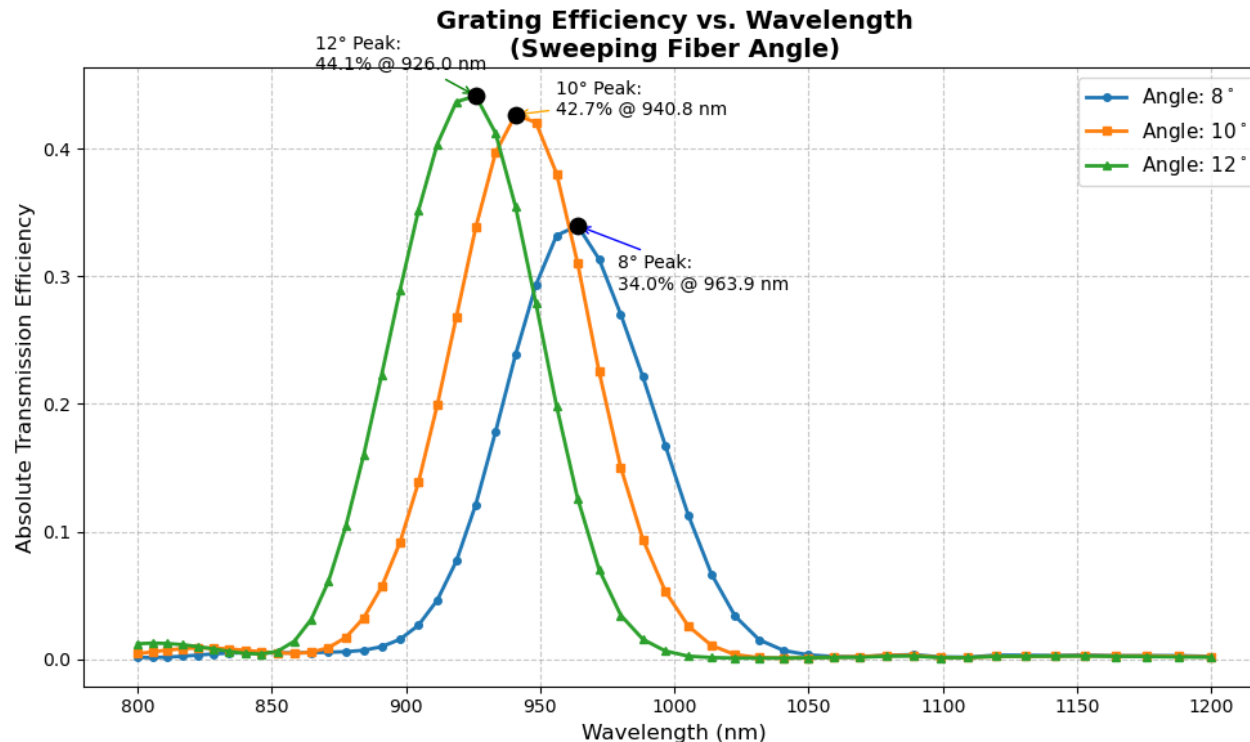
User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	0.25	um
3	[/]	etch depth	Length	0.18	um
4	[/]	duty cycle	Number	0.5	
5	[/]	pitch	Length	0.689	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	15	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[/]	index	Number	3.5	
11	[/]	material	Material	Si3N4 - 1000nm	
12	[/]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Fiber Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[/]	theta	Number	VARIABLE	
5	[/]	core index	Number	1.44427	
6	[/]	cladding index	Number	1.43482	



Increasing angle Increase the Efficiency

Peak wavelength shifted towards Left

Varying Grating Period

Etching Depth 0.180_{um}
Duty Cycle at 0.5
Fiber Angle 12
Box layer SiO₂ = 1.206_{um}

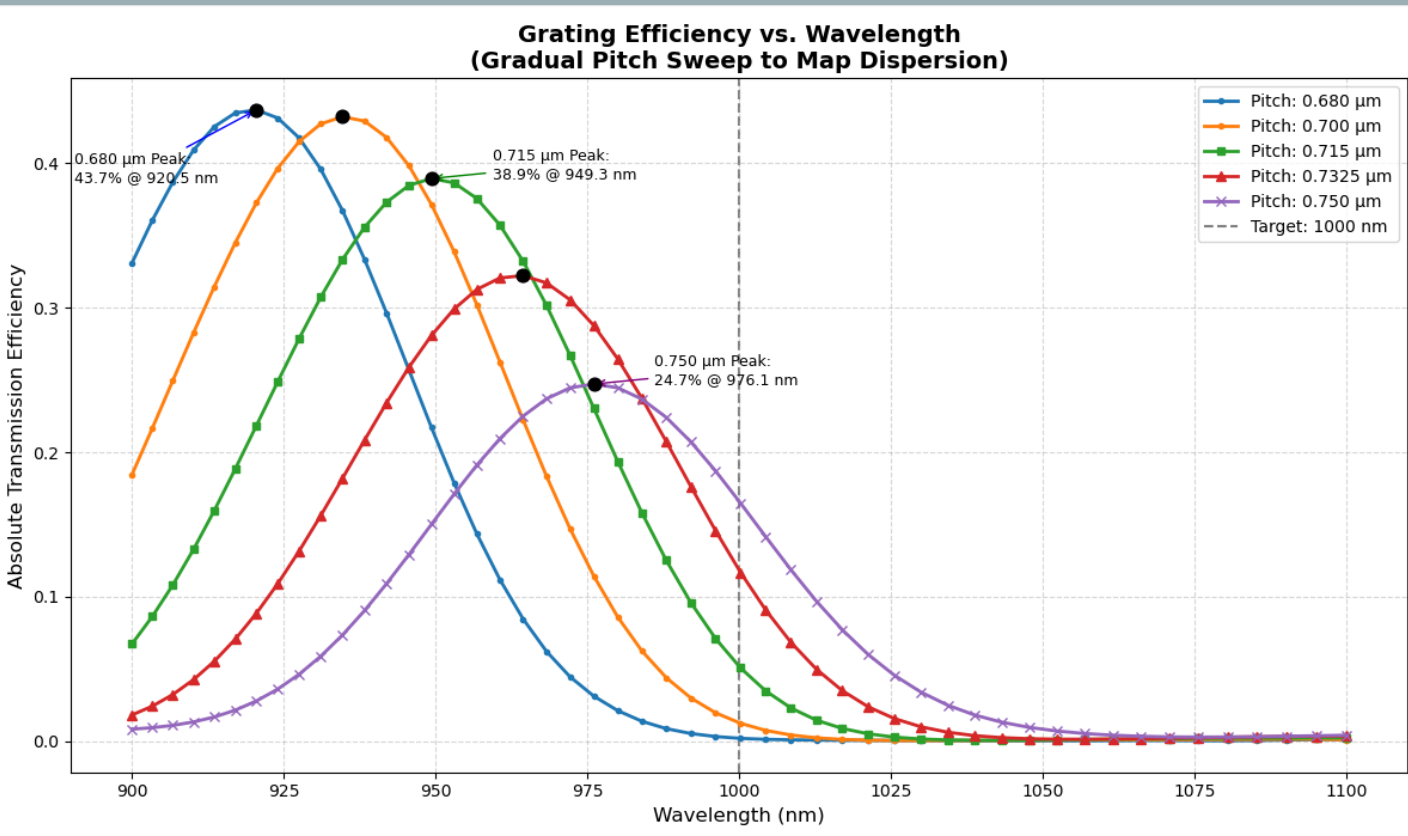
Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	0.25	um
3	[/]	etch depth	Length	0.18	um
4	[]	duty cycle	Number	0.5	
5	[/]	pitch	Length	VARIABLE	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	10	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	3.5	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Fiber Property

Increasing Grating Period Increase Efficiency
Peak wavelength shifted towards Left



Varying Grating Period

Etching Depth $0.30\text{ }\mu\text{m}$ Height of the Waveguide $0.35\text{ }\mu\text{m}$
Duty Cycle at 0.5
Fiber Angle 8
Box layer $\text{SiO}_2 = 0.817\text{ }\mu\text{m}$

Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	0.35	um
3	[/]	etch depth	Length	0.3	um
4	[]	duty cycle	Number	0.5	
5	[/]	pitch	Length	0.689	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	10	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	3.5	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

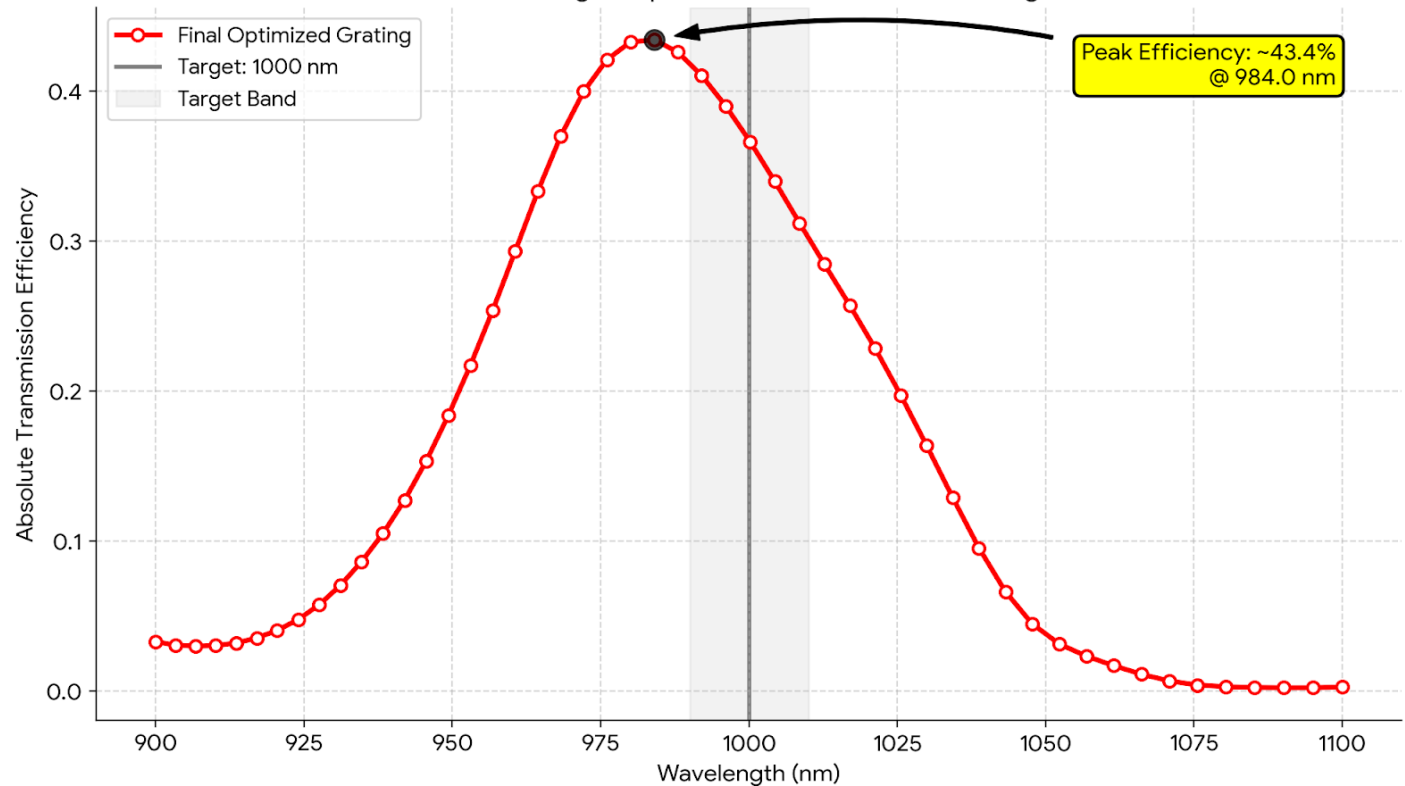
Fiber Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	8	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

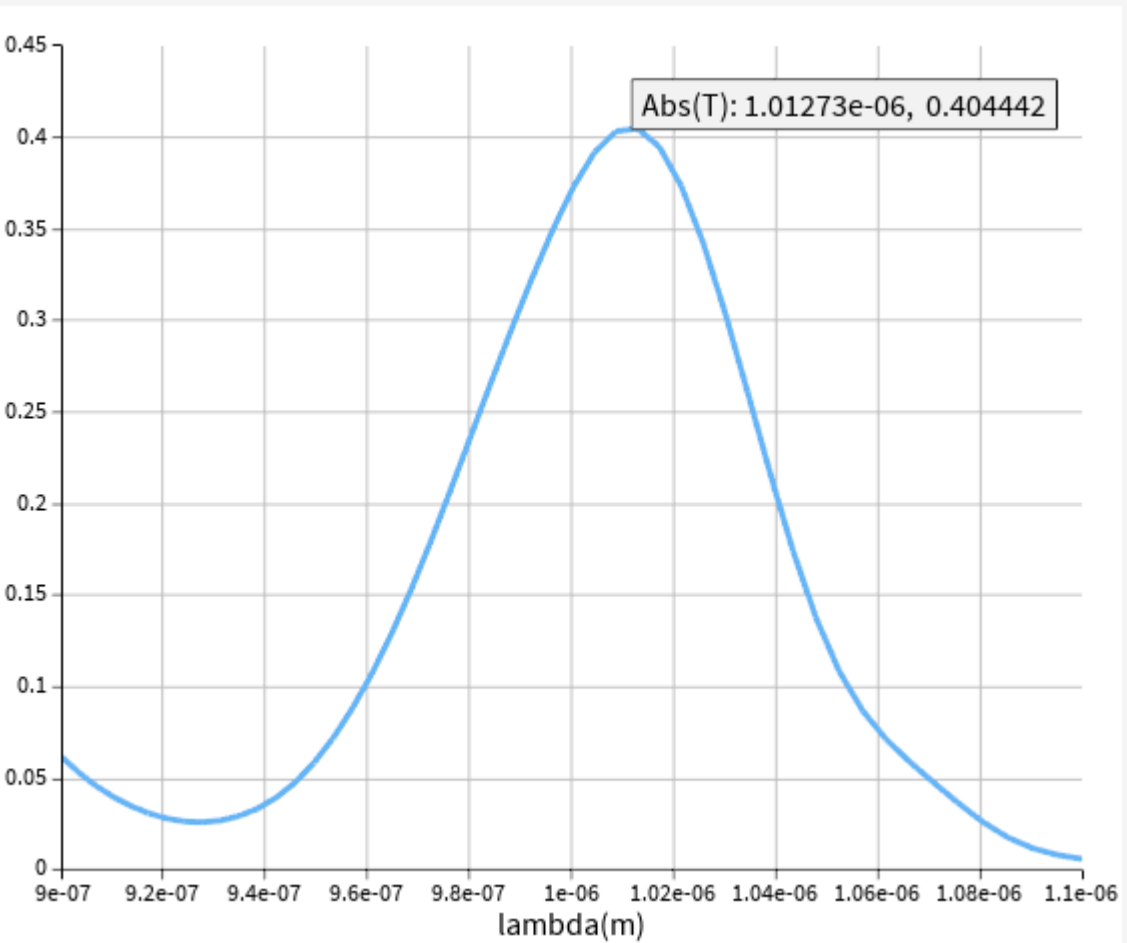
43.4% efficient at 984.0 nm

Final Grating Coupler Transmission vs. Wavelength



Varying Grating Period

Etching Depth 0.30_{um} Height of the Waveguide 0.4_{um}
Duty Cycle at 0.5
Fiber Angle 8
Box layer SiO₂ = 0.817_{um}



Grating Property

# ^	Name	Type	Value	Unit
1	[/] target length	Length	25	um
2	[/] h total	Length	0.4	um
3	[/] etch depth	Length	0.3	um
4	[] duty cycle	Number	0.5	
5	[/] pitch	Length	0.689	um
6	[/] radius	Length	25	um
7	[/] y span	Length	10	um
8	[/] L extra	Length	10	um
9	[/] waveguide width	Length	0.3	um
10	[] index	Number	3.5	
11	[] material	Material	Si3N4 - 1000nm	
12	[] m	Number	1.15	
13	[/] waveguide length	Length	10	um

Fiber Property

User properties

# ^	Name	Type	Value	Unit
1	[/] core diameter	Length	9	um
2	[/] cladding diameter	Length	50	um
3	[/] z span	Length	20	um
4	[] theta	Number	8	
5	[] core index	Number	1.44427	
6	[] cladding index	Number	1.43482	

40% efficient at 1000.0_{nm}

Varying Grating Period

Etching Depth $0.75\text{ }\mu\text{m}$
Duty Cycle at 0.5
Fiber Angle 8
Box layer $\text{SiO}_2 = 0.817\text{ }\mu\text{m}$
Height of the Waveguide $1.0\text{ }\mu\text{m}$

Grating Property

# ^	Name	Type	Value	Unit
1	[/] target length	Length	25	um
2	[/] h total	Length	1.0	um
3	[/] etch depth	Length	0.75	um
4	[] duty cycle	Number	0.5	
5	[/] pitch	Length	0.689	um
6	[/] radius	Length	25	um
7	[/] y span	Length	10	um
8	[/] L extra	Length	10	um
9	[/] waveguide width	Length	0.3	um
10	[] index	Number	3.5	
11	[] material	Material	Si3N4 - 1000nm	
12	[] m	Number	1.15	
13	[/] waveguide length	Length	10	um

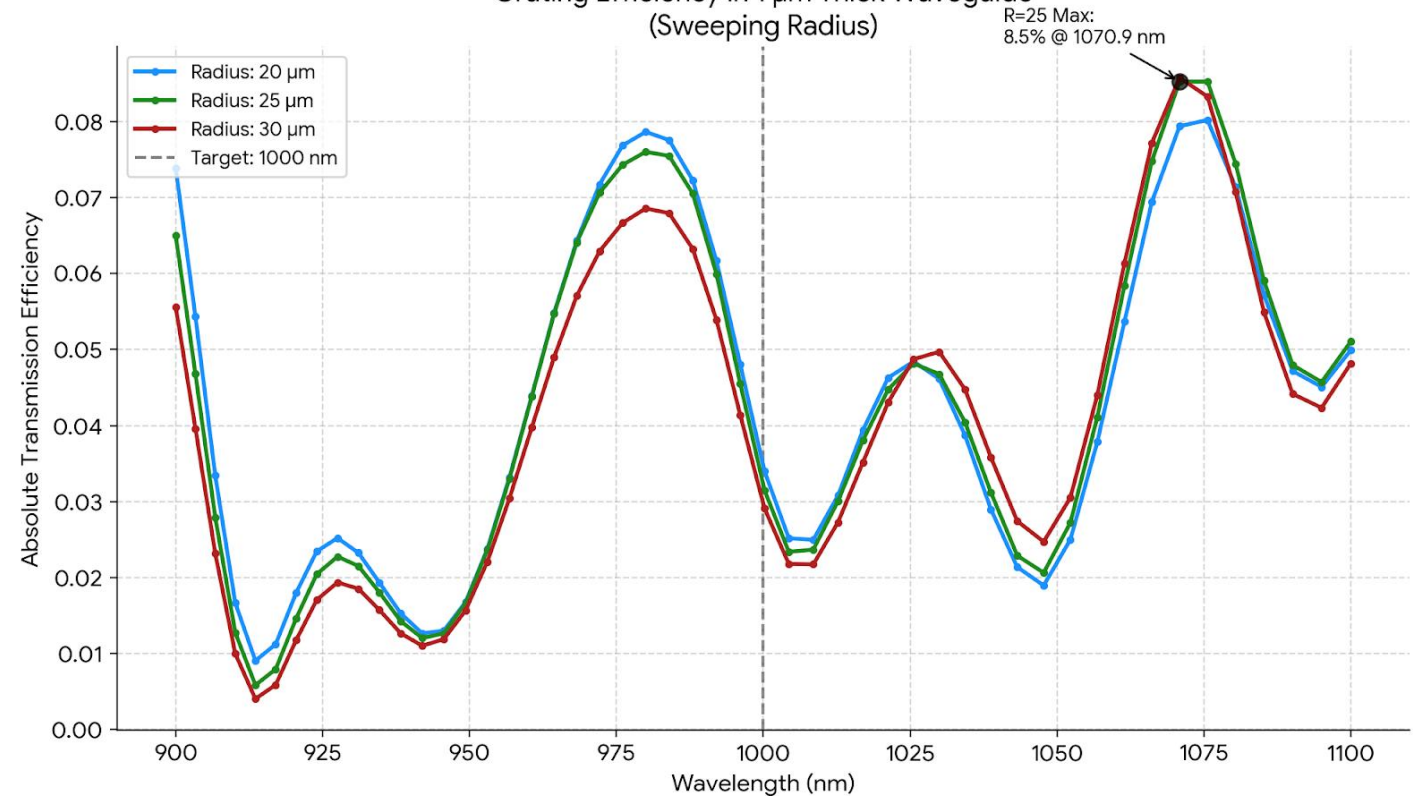
Fiber Property

User properties

# ^	Name	Type	Value	Unit
1	[/] core diameter	Length	9	um
2	[/] cladding diameter	Length	50	um
3	[/] z span	Length	20	um
4	[] theta	Number	8	
5	[] core index	Number	1.44427	
6	[] cladding index	Number	1.43482	

NOTHING MULTIPUL MODE EXIST

Grating Efficiency in 1 μm Thick Waveguide
(Sweeping Radius)



Varying Grating Period

Waveguide HIGHT 1.0 μm

Etch Depth: 0.150 μm

Duty Cycle: 0.5

Fiber Angle: 8°

Pitch: 0.550 μm

Box layer SiO_2 = 0.817 μm

Grating Property

User properties

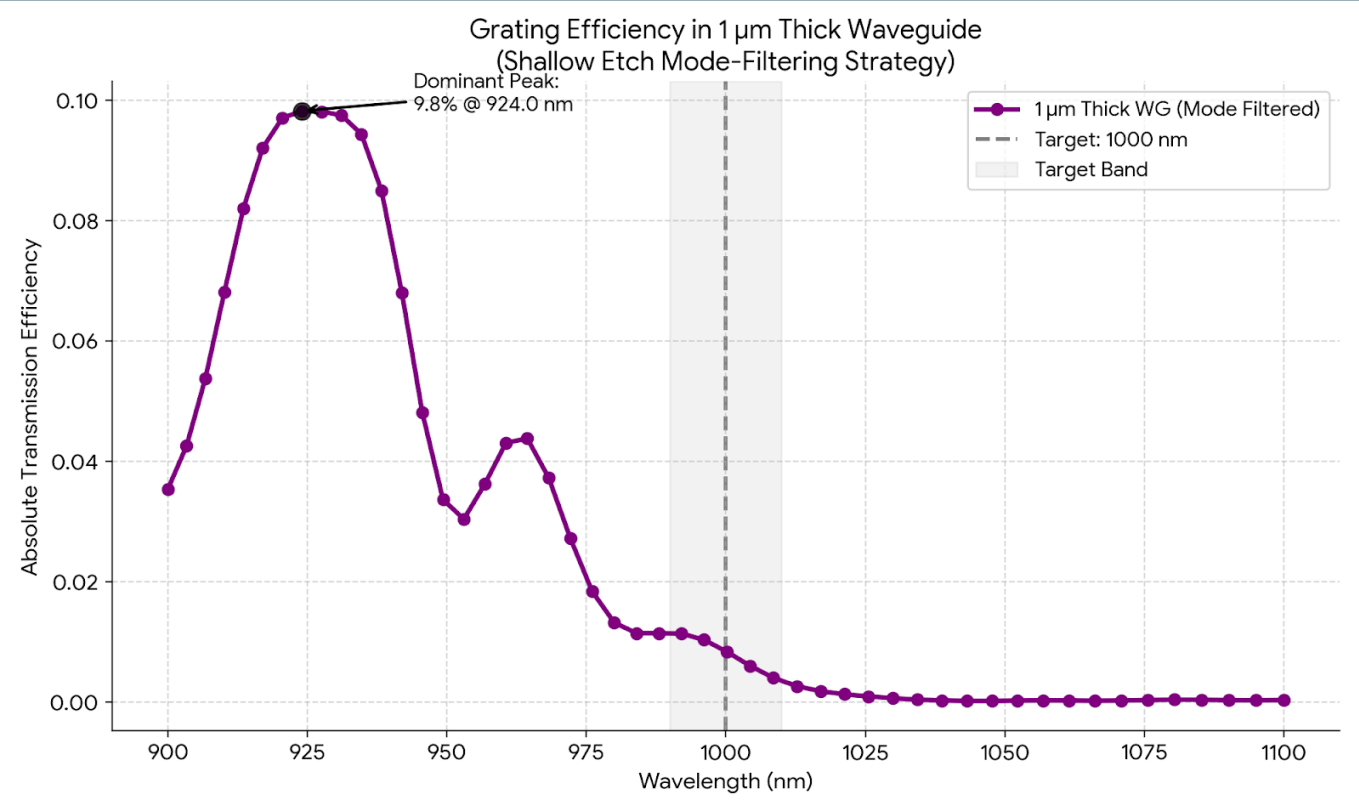
#	^	Name	Type	Value	Unit
1		target length	Length	25	um
2		h total	Length	1	um
3		etch depth	Length	0.15	um
4		duty cycle	Number	0.5	
5		pitch	Length	0.55	um
6		radius	Length	25	um
7		y span	Length	10	um
8		L extra	Length	10	um
9		waveguide width	Length	0.3	um
10		index	Number	3.5	
11		material	Material	Si3N4 - 1000nm	
12		m	Number	1.15	
13		waveguide length	Length	10	um

Fiber Property

User properties

#	^	Name	Type	Value	Unit
1		core diameter	Length	9	um
2		cladding diameter	Length	50	um
3		z span	Length	20	um
4		theta	Number	8	
5		core index	Number	1.44427	
6		cladding index	Number	1.43482	

40% efficient at 1000.0_{nm}



Varying Grating Period

Waveguide HIGHT 1.0 μm

Etch Depth: 150 nm

Duty Cycle: 0.5

Fiber Angle: 8°

Pitch: 0.60 μm

Box layer SiO₂ = 2.0 μm

pitch 0.6 DC 0.5 total Hight 1um etching 0.15um BOX layer 2um
Total Transmission vs. Wavelength

Grating Property

User properties

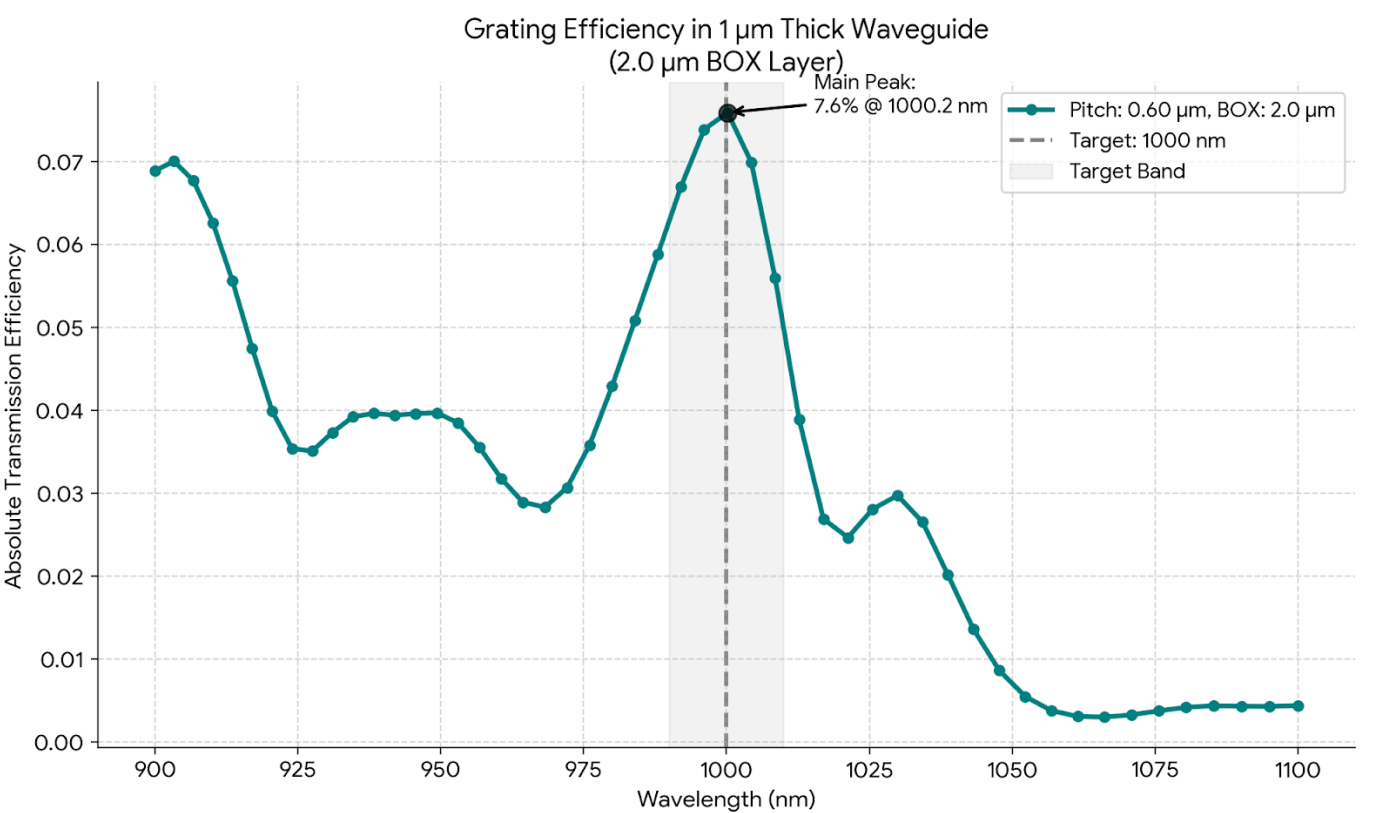
#	^	Name	Type	Value	Unit
1		target length	Length	25	um
2		h total	Length	1	um
3		etch depth	Length	0.15	um
4		duty cycle	Number	0.5	
5		pitch	Length	0.55	um
6		radius	Length	25	um
7		y span	Length	10	um
8		L extra	Length	10	um
9		waveguide width	Length	0.3	um
10		index	Number	3.5	
11		material	Material	Si3N4 - 1000nm	
12		m	Number	1.15	
13		waveguide length	Length	10	um

Fiber Property

User properties

#	^	Name	Type	Value	Unit
1		core diameter	Length	9	um
2		cladding diameter	Length	50	um
3		z span	Length	20	um
4		theta	Number	8	
5		core index	Number	1.44427	
6		cladding index	Number	1.43482	

40% efficient at 1000.0_{nm}



Varying Etch depth

Waveguide HIGHT 1.0 μm
Etch Depth: 0.1 to 0.5 μm
Duty Cycle: 0.5
Fiber Angle: 12°
Pitch: 0.70 μm
Box layer SiO₂ = 2.0 μm

Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	1	um
3	[/]	etch depth	Length	VARIABLE	um
4	[]	duty cycle	Number	0.5	
5	[/]	pitch	Length	0.7	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	10	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	3.5	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

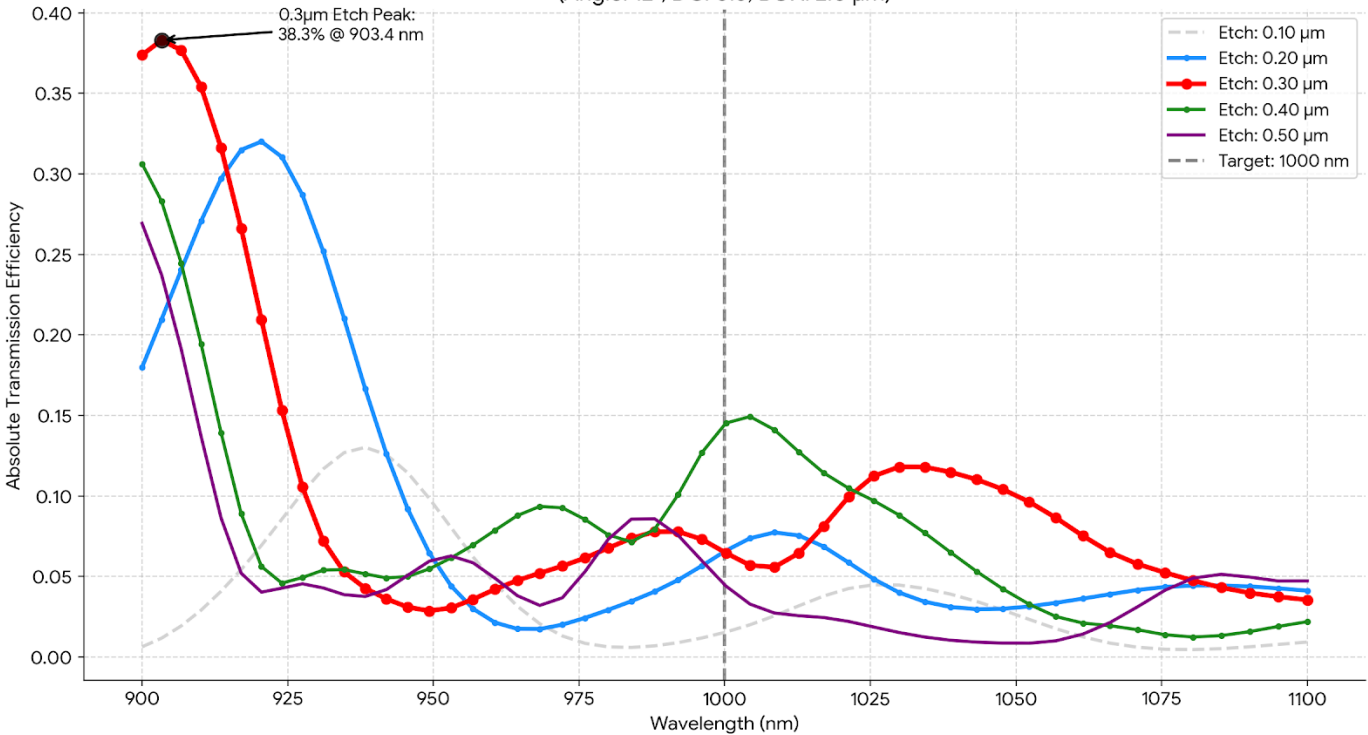
Fiber Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	12	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

38% efficient at 903.0_{nm}

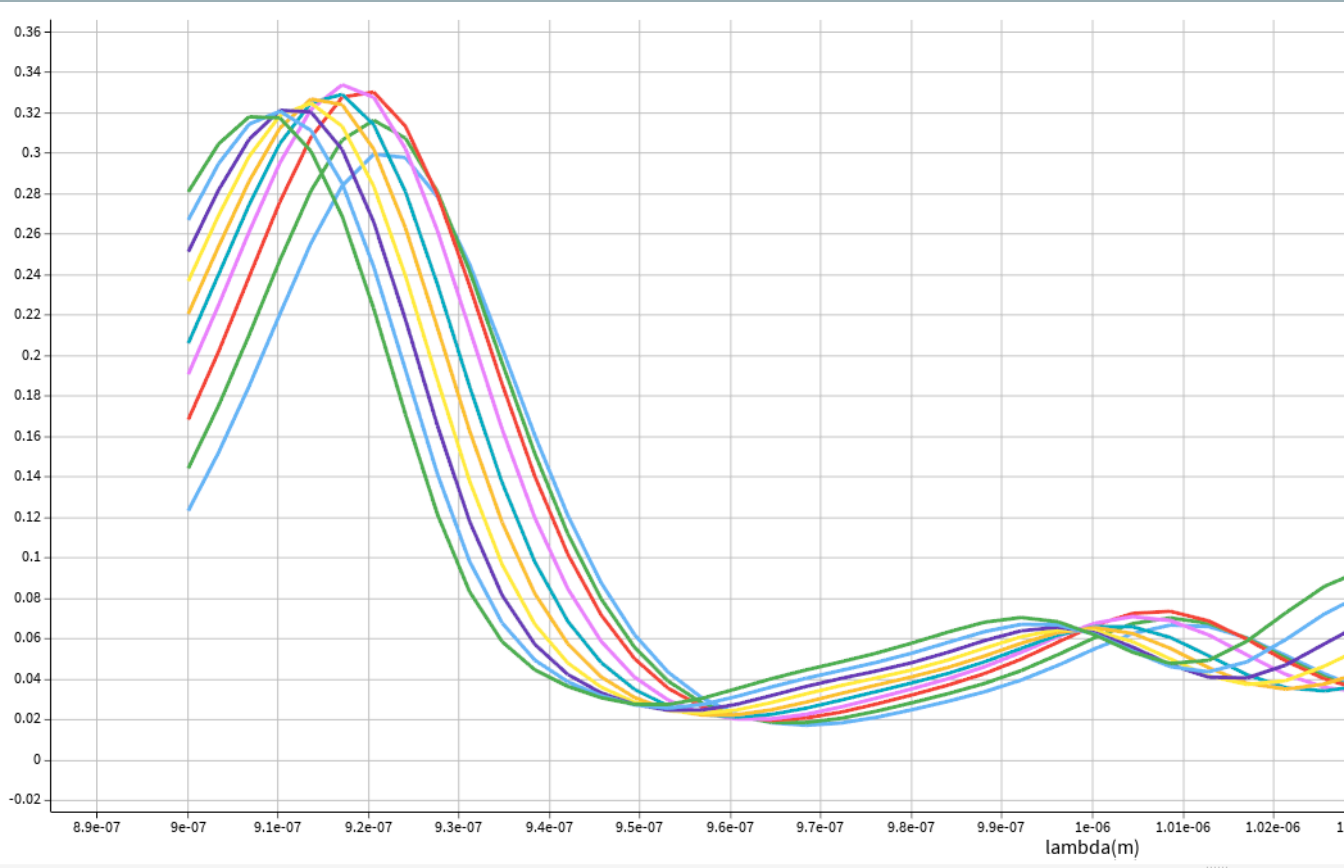
Etch Depth Sweep for 1 μm Thick Waveguide
(Angle: 12°, DC: 0.5, BOX: 2.0 μm)



Varying Etch depth

Waveguide HIGHT 1.0 μm
Etch Depth: 0.2 to 0.3 μm
Duty Cycle: 0.5
Fiber Angle: 12°
Pitch: 0.70 μm
Box layer SiO₂ = 2.0 μm

Pink line best at 0.22um etching depth



Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	1	um
3	[/]	etch depth	Length	VARIABLE	um
4	[]	duty cycle	Number	0.5	
5	[/]	pitch	Length	0.7	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	10	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	3.5	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Fiber Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	12	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

Varying Grating Period

TOTAL HEIGHT 1UM
BOX LAYER 2UM
ETCHING DEPTH 0.3UM
DUTY CYCLE 0.5
GRATING PERIOD FROM 0.65 TO 0.85UM
ANGLE 12

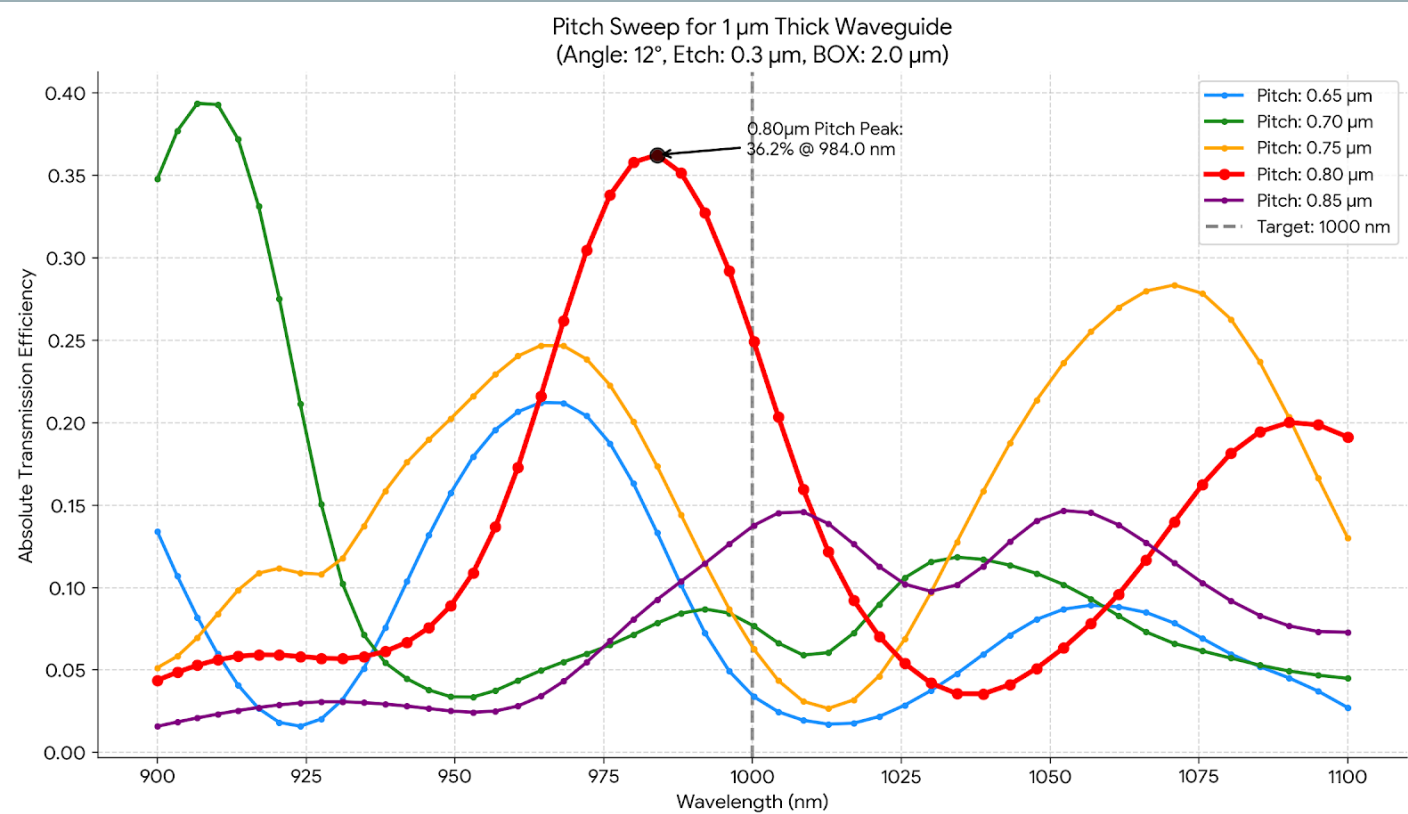
Grating Property

User properties				
# ^	Name	Type	Value	Unit
1	[/] target length	Length	25	um
2	[/] h total	Length	1	um
3	[/] etch depth	Length	0.3	um
4	[] duty cycle	Number	0.5	
5	[/] pitch	Length	VARIABLE	um
6	[/] radius	Length	25	um
7	[/] y span	Length	10	um
8	[/] L extra	Length	10	um
9	[/] waveguide width	Length	0.3	um
10	[] index	Number	3.5	
11	[] material	Material	Si3N4 - 1000nm	
12	[] m	Number	1.15	
13	[/] waveguide length	Length	10	um

Fiber Property

User properties				
# ^	Name	Type	Value	Unit
1	[/] core diameter	Length	9	um
2	[/] cladding diameter	Length	50	um
3	[/] z span	Length	20	um
4	[] theta	Number	12	
5	[] core index	Number	1.44427	
6	[] cladding index	Number	1.43482	

36% efficient at 984.0_{nm}



Varying Grating Period

TOTAL HEIGHT 1UM
BOX LAYER 2UM
ETCHING DEPTH 0.3UM
DUTY CYCLE 0.5
GRATING PERIOD FROM 0.81 TO 0.84UM
ANGLE 12

Grating Property

User properties

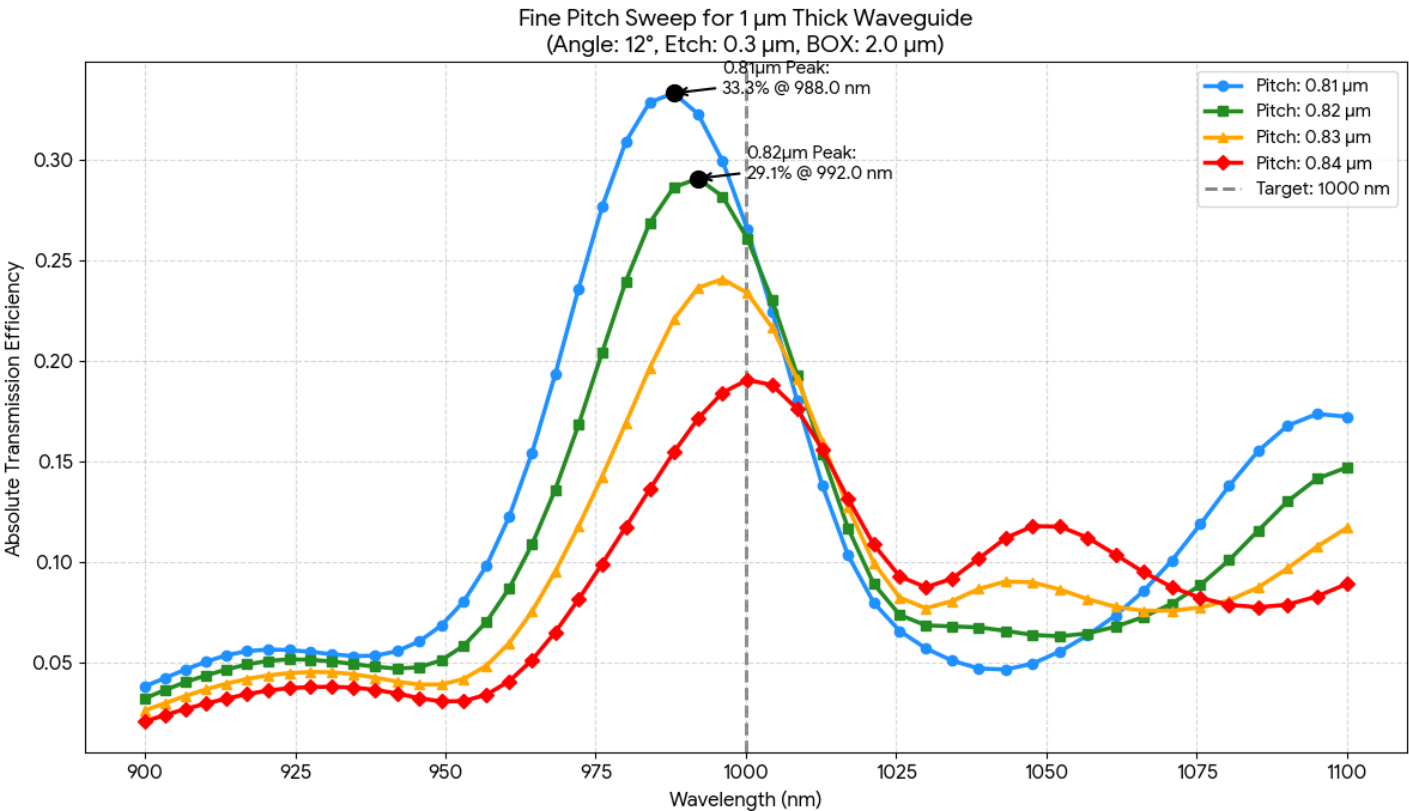
#	^	Name	Type	Value	Unit
1		target length	Length	25	um
2		h total	Length	1	um
3		etch depth	Length	0.3	um
4		duty cycle	Number	0.5	
5		pitch	Length	VARIABLE	um
6		radius	Length	25	um
7		y span	Length	10	um
8		L extra	Length	10	um
9		waveguide width	Length	0.3	um
10		index	Number	3.5	
11		material	Material	Si3N4 - 1000nm	
12		m	Number	1.15	
13		waveguide length	Length	10	um

Fiber Property

User properties

#	^	Name	Type	Value	Unit
1		core diameter	Length	9	um
2		cladding diameter	Length	50	um
3		z span	Length	20	um
4		theta	Number	12	
5		core index	Number	1.44427	
6		cladding index	Number	1.43482	

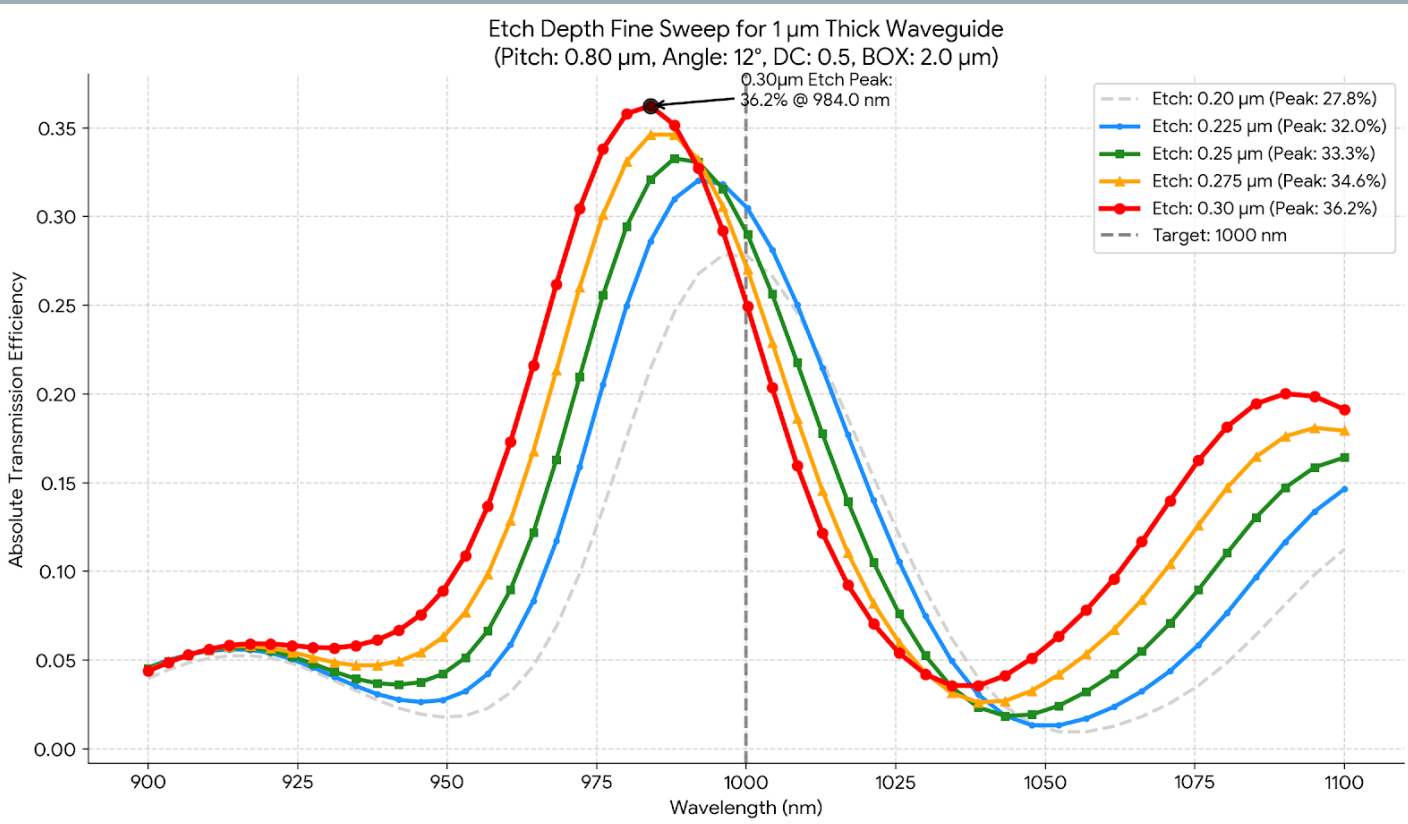
36% efficient at 984.0_{nm}



Varying Etch depth

Waveguide HIGHT 1.0 μm
Etch Depth: 0.2 to 0.3 μm
Duty Cycle: 0.5
Fiber Angle: 12°
Pitch: 0.80 μm
Box layer SiO₂ = 2.0 μm

0.225 IS 30% efficient at 1000.0_{nm}



Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	1	um
3	[/]	etch depth	Length	VARIABLE	um
4	[]	duty cycle	Number	0.5	
5	[/]	pitch	Length	0.8	um
6	[/]	radius	Length	25	um
7	[/]	y span	Length	10	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.3	um
10	[]	index	Number	3.5	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Fiber Property

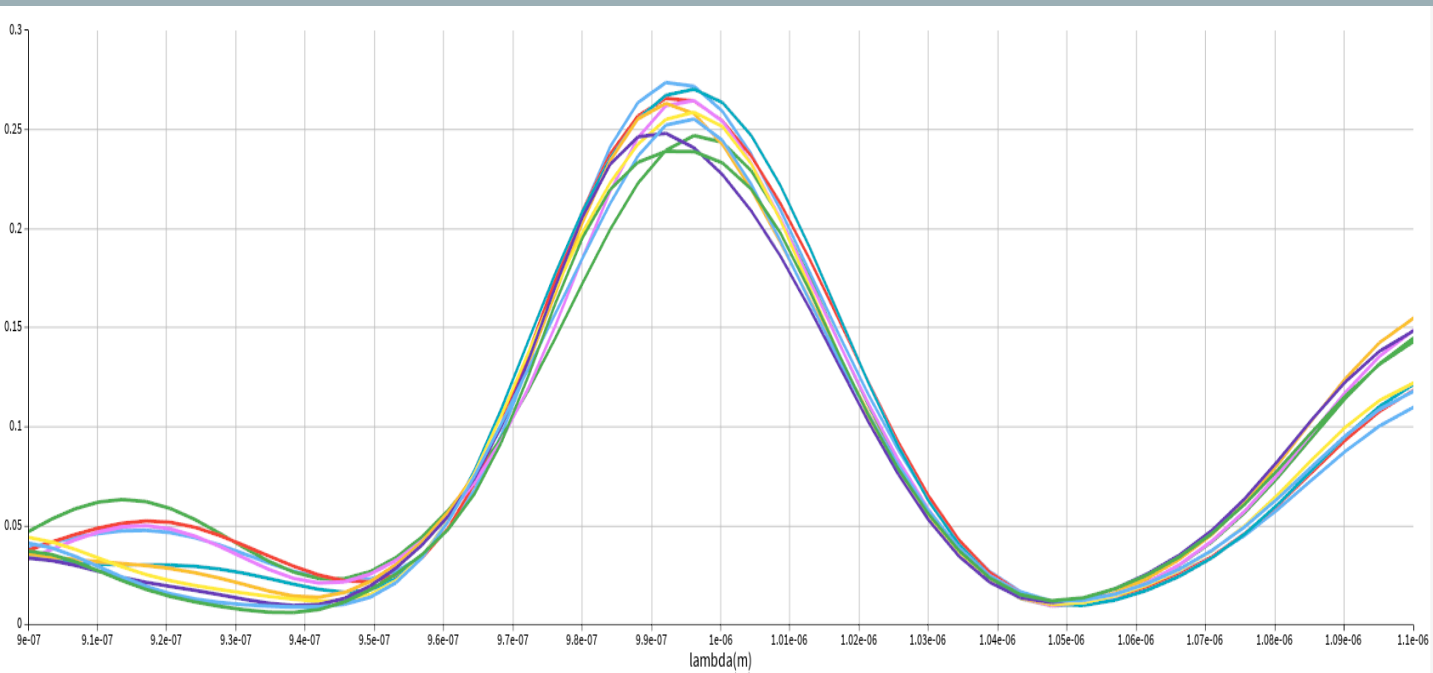
User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	12	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

Varying Taper length

Taper length 20 to 40 μm
Waveguide HIGHT 1.0 μm
Etch Depth: 0.225 μm
Duty Cycle: 0.5
Fiber Angle: 12°
Pitch: 0.80 μm
Box layer SiO_2 = 2.0 μm

BEST TAPER LENGTH 28.88UM



Grating Property

User properties				
# ^	Name	Type	Value	Unit
1	[/] target length	Length	25	um
2	[/] h total	Length	1	um
3	[/] etch depth	Length	0.225	um
4	[] duty cycle	Number	0.5	
5	[/] pitch	Length	0.8	um
6	[/] radius	Length	VARIABLE	um
7	[/] y span	Length	10	um
8	[/] L extra	Length	10	um
9	[/] waveguide width	Length	0.3	um
10	[] index	Number	3.5	
11	[] material	Material	Si3N4 - 1000nm	
12	[] m	Number	1.15	
13	[/] waveguide length	Length	10	um

Fiber Property

User properties				
# ^	Name	Type	Value	Unit
1	[/] core diameter	Length	9	um
2	[/] cladding diameter	Length	50	um
3	[/] z span	Length	20	um
4	[] theta	Number	12	
5	[] core index	Number	1.44427	
6	[] cladding index	Number	1.43482	

Varying WAVEGUIDE WIDTH

Taper length 28.88um

Waveguide HIGHT 1.0 μm

Etch Depth: 0.225 μm

Duty Cycle: 0.5

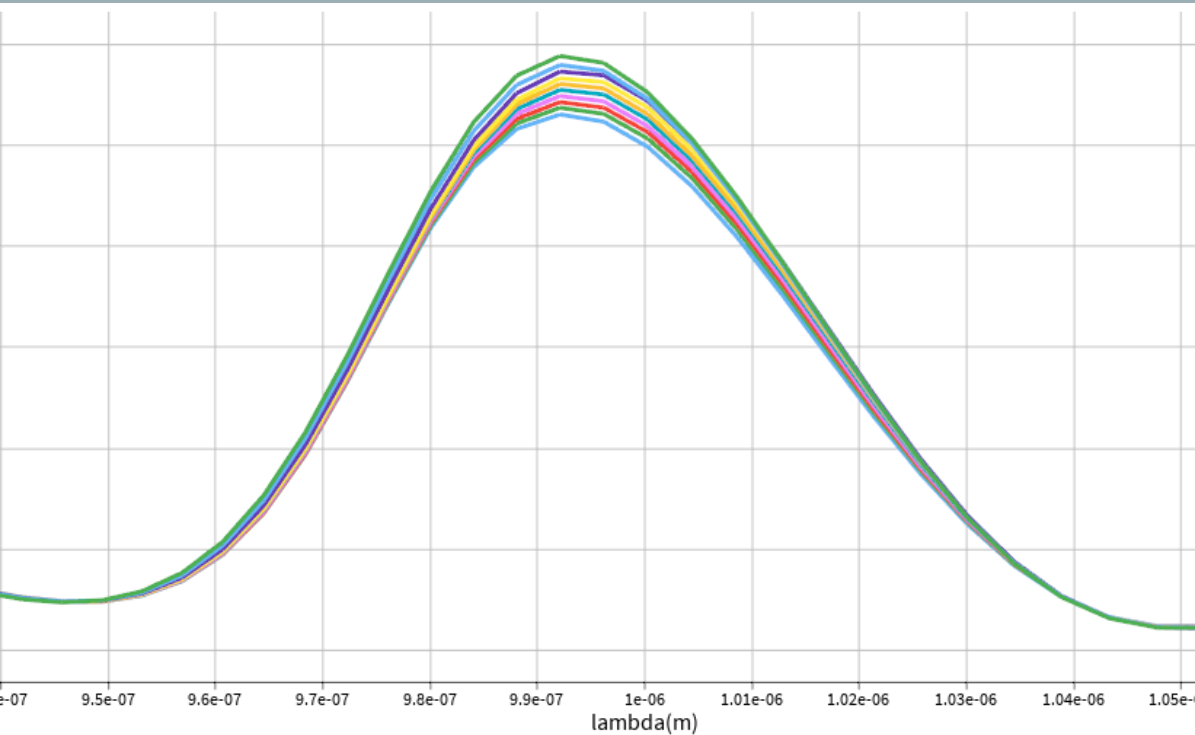
Fiber Angle: 12°

Pitch: 0.80 μm

Box layer SiO₂ = 2.0 μm

BEST WAVEGUIDE WIDTH 0.372 UM

INCREASING WAVEGUIDE WIDTH INCREASE TRANSMISSION



Grating Property

User properties				
# ^	Name	Type	Value	Unit
1	[/] target length	Length	25	um
2	[/] h total	Length	1	um
3	[/] etch depth	Length	0.225	um
4	[] duty cycle	Number	0.5	
5	[/] pitch	Length	0.8	um
6	[/] radius	Length	28.88	um
7	[/] y span	Length	10	um
8	[/] L extra	Length	10	um
9	[/] waveguide width	Length	VARIABLE	um
10	[] index	Number	3.5	
11	[] material	Material	Si3N4 - 1000nm	
12	[] m	Number	1.15	
13	[/] waveguide length	Length	10	um

Fiber Property

User properties				
# ^	Name	Type	Value	Unit
1	[/] core diameter	Length	9	um
2	[/] cladding diameter	Length	50	um
3	[/] z span	Length	20	um
4	[] theta	Number	12	
5	[] core index	Number	1.44427	
6	[] cladding index	Number	1.43482	

Varying DUTY CYCLE

Taper length 28.88 μm

Waveguide HIGHT 1.0 μm

Etch Depth: 0.225 μm

Duty Cycle: 0.45 TO 55

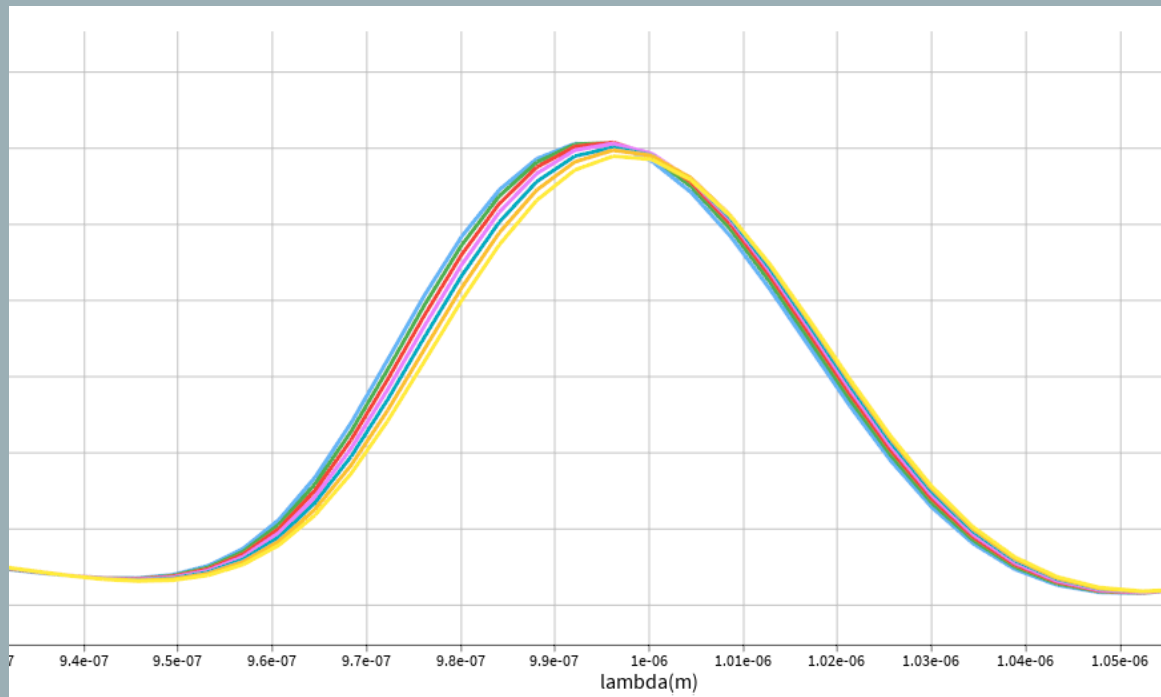
Fiber Angle: 12°

Pitch: 0.80 μm

Box layer SiO₂ = 2.0 μm

BEST DUTY CYCLE 0.51

NOT MUCH CHANGING FOR CHANGING DUTY CYCLE



Grating Property

User properties

# ^	Name	Type	Value	Unit
1	[/] target length	Length	25	um
2	[/] h total	Length	1	um
3	[/] etch depth	Length	0.225	um
4	[] duty cycle	Number	VARIABLE	
5	[/] pitch	Length	0.8	um
6	[/] radius	Length	28.88	um
7	[/] y span	Length	10	um
8	[/] L extra	Length	10	um
9	[/] waveguide width	Length	0.3	um
10	[] index	Number	3.5	
11	[] material	Material	Si3N4 - 1000nm	
12	[] m	Number	1.15	
13	[/] waveguide length	Length	10	um

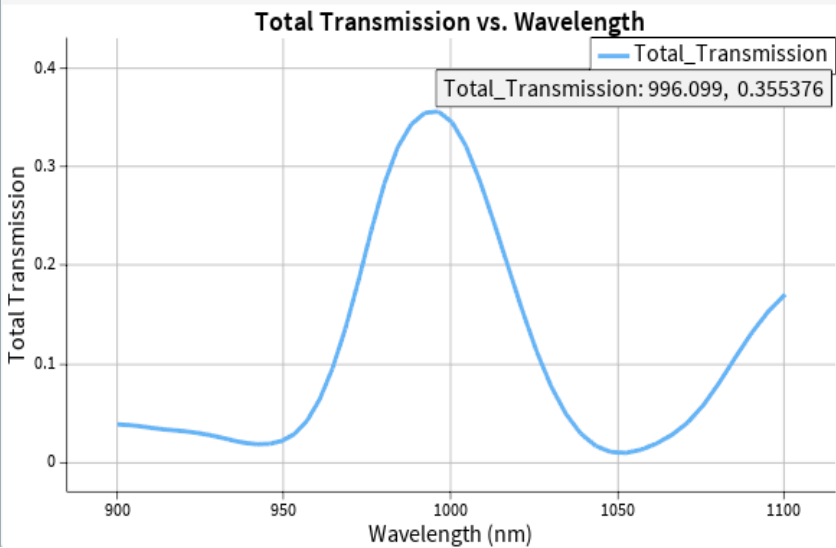
Fiber Property

User properties

# ^	Name	Type	Value	Unit
1	[/] core diameter	Length	9	um
2	[/] cladding diameter	Length	50	um
3	[/] z span	Length	20	um
4	[] theta	Number	12	
5	[] core index	Number	1.44427	
6	[] cladding index	Number	1.43482	

FINAL PARAMETERS

Taper length 28.88um
Waveguide HIGHT 1.0 μm
Waveguide width 0.372um
Etch Depth: 0.225 μm
Duty Cycle: 0.5
Fiber Angle: 12°
Pitch: 0.80 μm
Box layer SiO₂ = 2.0 μm



Grating Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	target length	Length	25	um
2	[/]	h total	Length	1	um
3	[/]	etch depth	Length	0.225	um
4	[]	duty cycle	Number	0.5	
5	[/]	pitch	Length	0.8	um
6	[/]	radius	Length	28.8	um
7	[/]	y span	Length	10	um
8	[/]	L extra	Length	10	um
9	[/]	waveguide width	Length	0.372	um
10	[]	index	Number	3.5	
11	[]	material	Material	Si3N4 - 1000nm	
12	[]	m	Number	1.15	
13	[/]	waveguide length	Length	10	um

Fiber Property

User properties

#	^	Name	Type	Value	Unit
1	[/]	core diameter	Length	9	um
2	[/]	cladding diameter	Length	50	um
3	[/]	z span	Length	20	um
4	[]	theta	Number	12	
5	[]	core index	Number	1.44427	
6	[]	cladding index	Number	1.43482	

